

Research Specification Document

Scenario-Aware Health Indicator Efficacy Analysis for LFP Battery SOH Estimation

Idea 1 — Improved Version
Based on iTransformer Architecture

March 2026

1. Executive Summary

This document specifies the complete research pipeline for a systematic analysis of Health Indicator (HI) efficacy across diverse partial-discharge scenarios in LFP batteries. The study uses the iTransformer architecture both as a prediction model and as an analytical tool for uncovering multivariate feature interactions via cross-variate attention weights.

Core Research Question: Does the optimal set of Health Indicators change depending on the partial-discharge scenario, and if so, how?

Key Outputs: (1) HI Efficacy Map — a scenario $\times$ HI heatmap quantifying each indicator's contribution per scenario; (2) Robust vs. Scenario-Specific HI classification — practical guidelines for BMS engineers; (3) Supervised vs. Unsupervised feature selection gap analysis — quantifying real-world deployability; (4) Cross-variate attention-based multivariate interaction networks per scenario.

Target Journals: Applied Energy, Journal of Power Sources, IEEE Transactions on Transportation Electrification

Estimated Timeline: 12–16 weeks from data preprocessing to manuscript submission

2. Research Background and Motivation

2.1 Problem Statement

Data-driven SOH estimation models are typically trained and validated under fixed laboratory conditions (constant-current full discharge, controlled temperature). In real-world EV applications, batteries experience partial discharges across varying SOC ranges, current profiles, and temperatures. The Health Indicators that are optimal under lab conditions may not remain optimal under these diverse real-world scenarios.

Existing HI selection studies (Tao et al., J. Energy Storage 2024; Li et al., J. Power Sources 2021) perform feature selection under a single fixed condition and produce a single optimal HI subset. No prior work has systematically investigated how HI efficacy varies across different partial-discharge scenarios or provided scenario-dependent guidelines.

2.2 Novelty and Differentiation

Novelty 1: First systematic cross-scenario HI efficacy analysis. The question 'Does the optimal HI set change across scenarios?' has not been formally posed in the literature.

Novelty 2: Supervised vs. unsupervised feature selection gap quantification per scenario. This directly addresses real-world deployability where SOH labels are unavailable.

Novelty 3: Use of iTransformer cross-variate attention weights as a multivariate HI interaction analysis tool, surpassing traditional univariate correlation-based HI evaluation.

Novelty 4: HI Efficacy Map as a reusable community resource — a lookup table for BMS engineers to select HIs based on their specific operating profile.

2.3 Related Work and Positioning

Study	Method	Limitation	Our Difference
Tao et al. (2024)	61 HIs + MI + Bi-LSTM	Single condition, univariate selection	Multi-scenario, multivariate attention
Li et al. (2021)	Partial charge curve HI analysis	Single variable sensitivity only	Systematic scenario classification
Deng et al. (2022)	Random charge segment + SGPR/DCNN	Single 'random' setting, no scenario taxonomy	Taxonomy of scenarios + cross-comparison
iTransformer SOH (2025)	TCN + enhanced iTransformer	Fixed feature system, no scenario analysis	Scenario-adaptive analysis framework

3. Dataset Specification

3.1 Primary Datasets

Dataset	Cells	Chemistry	Charge Condition	Discharge Condition
MIT-Stanford	124	A123 LFP	72 fast-charge protocols (3.6C–6C)	Uniform 4C CC
HUST	77	A123 LFP	Uniform 0.5C CC-CV	77 multi-stage protocols

Rationale for dataset pairing: Both datasets use identical A123 LFP cells, controlling chemistry as a variable. MIT provides charging-condition diversity with uniform discharge; HUST provides discharge-condition diversity with uniform charging. This complementarity enables controlled comparisons.

3.2 Data Access

MIT-Stanford: <https://data.matr.io/1> (public, Severson et al. Nature Energy 2019)

HUST: <https://data.mendeley.com/datasets/nsc7hnsg4s/2> (public)

3.3 Supplementary Datasets (for Extended Validation)

Dataset	Cells	Chemistry	Usage
SNL (Sandia)	86	NCA/NMC/LFP	Cross-chemistry generalization test
CALCE	Multiple	LCO/LFP/NMC	Dynamic drive profile (DST, FUDS) validation

4. Phase A — Data Preprocessing

4.1 Step A-1: Scenario Classification [Priority: ★★★★★]

This is the foundational step. All subsequent analyses depend on the quality and defensibility of scenario definitions.

4.1.1 Manual Scenario Classification (Physics-Based)

Applied to BOTH MIT and HUST by slicing full-discharge curves at predefined SOC ranges:

Scenario	SOC Range	Δ SOC	Real-World Analogy
H	80% → 60%	20%	Urban commute (frequent partial use in high SOC)
L	40% → 10%	30%	Long-trip tail end (deep discharge from low SOC)
W	90% → 20%	70%	Highway continuous driving (wide range)
R	Random start/end	10–50%	Irregular usage (random SOC window per cycle)
F	100% → 0%	100%	Full discharge baseline (lab reference)

Implementation for MIT: Each cell's full 4C discharge curve (SOC 100%→0%) is sliced at the specified SOC ranges. For Scenario R, start SOC is sampled from $U[20,90]$ and Δ SOC from $U[10,50]$ per cycle. The full curve remains available as ground truth.

Implementation for HUST: Each cell's multi-stage discharge curve is similarly sliced at the same SOC ranges. Since HUST protocols already have varying current profiles, each scenario (e.g., H) will contain data with diverse current patterns — this enables the additional 'current-pattern effect' sub-analysis unique to HUST.

4.1.2 Automatic Scenario Classification (Data-Driven)

Applied to BOTH datasets using the same pipeline to ensure unified scenario definitions.

Step 1 — Generate random partial observations.

From each cycle's full-discharge curve, extract multiple partial segments by uniformly sampling: $\text{start_SOC} \sim U[10, 95]$, $\Delta\text{SOC} \sim U[10, 80]$, $\text{end_SOC} = \max(0, \text{start_SOC} - \Delta\text{SOC})$. This generates 5–10 partial observations per cycle across all cells, yielding ~500K (MIT) + ~700K (HUST) samples.

Step 2 — Extract scenario descriptors (NOT Health Indicators).

For each partial observation, compute the following descriptor vector $d \in \mathbb{R}^5$:

- start_SOC : SOC at the beginning of the observed segment
- end_SOC : SOC at the end of the observed segment
- ΔSOC : width of the observed SOC window ($= \text{start_SOC} - \text{end_SOC}$)
- mean_V : average terminal voltage within the segment
- range_V : voltage span ($V_{\text{max}} - V_{\text{min}}$) within the segment

- mean_I: average discharge current within the segment
- std_I: standard deviation of current (≈ 0 for MIT, >0 for HUST)
- duration: total time of the observed segment (seconds)

Step 3 — Normalize and cluster.

Apply z-score normalization to all descriptors. Run K-means ($K=2..10$) on the combined MIT+HUST descriptor matrix. Select optimal K via silhouette score maximization. Additionally run GMM with BIC minimization as a secondary method. Compare both results.

Step 4 — Interpret and validate clusters.

Analyze each cluster centroid to assign physical meaning. Compare automatic clusters with manual scenarios (H/L/W/R) using a confusion matrix. Visualize using t-SNE/UMAP 2D projection and radar charts per cluster.

Key design decision: Since MIT has uniform current ($\text{std_I} \approx 0$), clusters in MIT will be driven primarily by SOC-range descriptors. In HUST, both SOC-range and current-pattern descriptors contribute. Clustering both datasets together ensures a common scenario space.

4.2 Step A-2: SOC Overlap Identification and Feature Extraction [Priority: ★★★★★]

4.2.1 SOC Overlap Search

Within each scenario, select K consecutive cycles from the same cell. Identify the common SOC sub-range observed across all K cycles (the overlap region). This overlap has length L (in SOC %).

Parameters: $K \in \{5, 10, 20\}$ (sliding window length). Stride $\in \{1, 3, 5\}$ cycles. Minimum overlap threshold: $L_{\min} = 5\%$ SOC. If overlap $< L_{\min}$, discard that window.

Note on Scenario F (full discharge): Overlap is always 100%, serving as the upper-bound reference for all partial scenarios.

4.2.2 Feature Pool Construction ($M+\alpha$ Features)

Extract $M+\alpha = 30$ candidate features per sliding window from the overlap region. Features are organized by physical dimension following Chen et al. (Renewable and Sustainable Energy Reviews, 2025):

Electrical features (12):

- Mean discharge voltage, voltage standard deviation, voltage skewness, voltage kurtosis
- Mean dV/dt (voltage slope), std of dV/dt
- $\Delta Q(V)$ variance (Severson et al. method, adapted to overlap region)
- Discharge capacity within overlap ($\int I dt$)
- IC peak height and position (if detectable in the overlap range)
- Voltage at fixed SOC points within overlap (2–3 anchor points)

Thermal features (5):

- Mean temperature, max temperature, temperature rise rate (dT/dt)
- Max–min temperature difference, temperature entropy

Temporal features (5):

- Time to traverse the overlap SOC range
- Time in specific voltage sub-bands (2 sub-bands within overlap)
- Equal-voltage dwell time ratio, discharge rate ($\Delta SOC / \text{time}$)

Cross-cycle features (8):

- Cycle-to-cycle difference (Δ) of each of the above 4 key features
- Moving average (window=5) and moving std of each of the above 4 key features

Important: Feature pool is identical across all scenarios. The question is which subset is most effective per scenario, not whether different features are available.

4.3 Step A-3: Feature Subset Construction [Priority: ★★★★★]

Construct multiple HI subsets along three axes. Each axis independently generates subsets.

4.3.1 Axis 1 — Selection Method (6 methods)**Supervised methods (SOH label available):**

- Pearson correlation: top-M features by $|\rho(HI, SOH)|$
- Spearman rank correlation: top-M features by $|\rho_s(HI, SOH)|$
- mRMR (Maximum Relevance Minimum Redundancy): MI-based, balances SOH-relevance and inter-feature redundancy

Unsupervised methods (SOH label NOT used):

- Variance-based + redundancy removal: remove features with correlation > 0.95 among themselves, then keep top-M by variance
- PCA-based: select M original features with highest loadings on top principal components
- Laplacian Score: select features that best preserve local data geometry

4.3.2 Axis 2 — Feature Category (5 groups)

- V-only: electrical features only (12→M)
- T-only: thermal features only (5→M, may be $<M$)
- Time-only: temporal features only (5→M, may be $<M$)
- V+T: electrical + thermal combined
- All: full pool (30→M)

4.3.3 Axis 3 — Subset Size (5 sizes)

- $M = 5, 10, 15, 20, \text{All}(30)$

Total core experiments: 6 selection methods $\times$ 4 manual scenarios (H/L/W/R) + F baseline = 30 runs per model. Category and size axes are ablation studies.

4.4 Step A-4: Data Splitting [Priority: ★★★★★]

Absolute Rule: Cell-level split. No single cell appears in more than one of train/val/test. This prevents information leakage.

Dataset	Train	Validation	Test
MIT (124 cells)	70 cells (56%)	24 cells (19%)	30 cells (25%)
HUST (77 cells)	44 cells (57%)	15 cells (19%)	18 cells (24%)

Stratification: Ensure that the distribution of charging protocols (MIT) or discharge protocols (HUST) is approximately balanced across train/val/test splits. Use stratified random sampling with protocol-type as the stratification variable.

Cross-dataset protocol: For generalizability testing, train on MIT → test on HUST, and vice versa. The same scenario definitions (H/L/W/R or automatic clusters) are applied to both datasets.

5. Phase B — Model Architecture

5.1 Primary Model: iTransformer + MLP [Priority: ★★★★★]

5.1.1 Architecture

Input: M features $\times$ K cycles matrix (one sample). Each of the M features is treated as an independent token of length K (inverted tokenization — features as tokens, not time steps).

Token embedding: Each M -dimensional feature's K -length time series is projected to d_{model} dimensions via linear projection. $d_{\text{model}} \in \{64, 128\}$.

Transformer encoder: $L = 2\text{--}3$ layers. Multi-head cross-variate self-attention (heads = $4\text{--}8$) computes attention across the M feature tokens, learning which features attend to which other features. Feed-forward network applied per-token independently.

Positional encoding: Applied along the cycle axis (K dimension) to encode degradation temporal order.

SOH regression head: Global average pooling over M tokens $\rightarrow$ 2-layer MLP ($d_{\text{model}} \rightarrow 32 \rightarrow 1$) $\rightarrow$ sigmoid for $[0, 1]$ range.

Estimated parameters: $\sim 15\text{K--}30\text{K}$ depending on d_{model} and M . Lightweight enough for BMS edge deployment.

5.1.2 Attention Weight Extraction Pipeline

After training, extract the $M \times M$ attention weight matrix from each layer and head. Average across all layers and heads to produce a single $M \times M$ matrix A per scenario. $A[i, j]$ represents 'how much feature i attends to feature j '. Row-sum $A[i, :]$ indicates feature i 's 'broadcaster' importance; column-sum $A[:, j]$ indicates feature j 's 'receiver' importance. The column-sum is used as the multivariate importance score for HI ranking.

5.2 Baseline Model 1: Random Forest [Priority: ★★★★★]

Input: Same $M \times K$ matrix, flattened to $M \cdot K$ dimensional vector.

Configuration: $n_{\text{estimators}}=500$, max_depth tuned via Bayesian optimization on validation set.

Feature importance: RF provides built-in Gini importance per feature. Compare RF feature importance ranking with iTransformer attention-based ranking per scenario. Agreement between two fundamentally different models strengthens findings; disagreement reveals model-feature interactions.

5.3 Baseline Model 2: LSTM [Priority: ★★★★★]

Input: $M \times K$ matrix, fed as a sequence of K time steps, each with M features.

Configuration: 2-layer LSTM, $\text{hidden_size}=64$. Last hidden state $\rightarrow$ MLP $\rightarrow$ SOH.

Role: Standard sequential baseline. Required by reviewers. Less analytical value than RF but essential for completeness.

5.4 Training Configuration

Parameter	Value
Loss function	MSE (Mean Squared Error)
Optimizer	AdamW (weight_decay=1e-4)
Learning rate	1e-3 with cosine annealing to 1e-5
Batch size	64
Epochs	200 with early stopping (patience=20 on val RMSE)
Framework	PyTorch 2.x
Hardware	Single GPU (RTX 3090 or equivalent)

6. Phase C — Experiments and Analysis

6.1 Result Matrix Construction [Priority: ★★★★★]

The central output is a 3D performance tensor: Scenario × HI Subset × Model.

Primary matrix (per model): Rows = scenarios {H, L, W, R, F, Auto-1, Auto-2, ...}. Columns = HI subsets {Pearson-M, Spearman-M, mRMR-M, Variance-M, PCA-M, Laplacian-M}. Cell value = test-set RMSE (%).

Three such matrices are produced (iTransformer, RF, LSTM). Additional matrices for category-axis and size-axis ablations.

6.2 Analysis 1 — Optimal Subset per Scenario [Priority: ★★★★★]

For each scenario row, identify the column (HI subset) with lowest RMSE. If the same subset is optimal across all scenarios, conclude that a universal HI set suffices. If different subsets are optimal for different scenarios, the core thesis (scenario-dependence of HI efficacy) is confirmed.

Cross-model consistency check: Verify whether the optimal-subset ranking is consistent across iTransformer, RF, and LSTM. If consistent, the finding is model-agnostic (stronger). If inconsistent, model-feature interaction exists (a secondary finding).

6.3 Analysis 2 — Robust vs. Scenario-Specific HI [Priority: ★★★★★]

Rank all $M+\alpha$ individual HIs by their average contribution across all scenarios (using RF Gini importance or iTransformer attention column-sum). HIs that rank in the top 20% across ALL scenarios are classified as 'Robust HI'. HIs that rank in the top 20% in ONLY specific scenarios are classified as 'Scenario-Specific HI'. Present as a heatmap (the HI Efficacy Map): rows = individual HIs, columns = scenarios, cell color = importance score.

Practical output: A BMS engineer can look up their operating profile (e.g., mostly urban commute = Scenario H) and immediately see which HIs to prioritize.

6.4 Analysis 3 — Supervised vs. Unsupervised Gap [Priority: ★★★★★☆]

For each scenario, compute: $\text{Gap}(s) = \text{RMSE_unsupervised}(s) - \text{RMSE_supervised}(s)$. Plot Gap across scenarios. Scenarios where $\text{Gap} < 1\%$ indicate that unsupervised selection is practically sufficient (no SOH label needed). Scenarios where $\text{Gap} > 3\%$ indicate that SOH labels are critical for that operating condition.

Practical output: Guidance on which real-world scenarios can rely on label-free feature selection versus which require periodic calibration measurements.

6.5 Analysis 4 — Attention-Based Interaction Networks [Priority: ★★★★★☆]

Visualize the $M \times M$ attention matrix as a network graph per scenario. Nodes = HIs, edge thickness = attention weight. Compare network topology across scenarios. Key questions: Do high-SOC scenarios show tighter voltage-feature clustering? Do variable-current scenarios show stronger temperature-voltage cross-edges?

Physical interpretation required: Each observed pattern must be linked to an electrochemical mechanism. E.g., 'In Scenario L, temperature HIs gain importance because LFP's flat voltage plateau in mid-SOC provides minimal voltage-based information, forcing the model to rely on thermal signatures of internal resistance increase.'

6.6 Analysis 5 — Subset Size Saturation [Priority: ★★★★★☆]

For each scenario, plot RMSE vs. M ($5 \rightarrow 10 \rightarrow 15 \rightarrow 20 \rightarrow 30$). Identify the saturation point (elbow) where additional features yield diminishing returns. Report as lightweight guidance: 'In Scenario H, 10 features achieve 95% of full-pool performance.'

Placement: Main text if saturation points differ significantly across scenarios (reinforcing core thesis); otherwise in supplementary materials.

6.7 Analysis 6 — Cross-Dataset Validation [Priority: ★★★★★★]

Train on MIT using Scenario H's optimal HI subset $\rightarrow$ test on HUST Scenario H. And vice versa. Performance relative to within-dataset results quantifies generalizability.

HUST-only sub-analysis: Within HUST Scenario H, compare cells from high C-rate protocols vs. low C-rate multi-stage protocols. Since SOC range is controlled (both are H), any performance difference is attributable to current-pattern effects. This analysis is impossible with MIT (uniform 4C) and represents a unique HUST contribution.

Analysis	Priority	Core Question	Success Metric	Fallback if Weak
1. Optimal subset	★★★★★	Does optimal HI set vary by scenario?	Different subsets win in different rows	Universal set also a valid finding
2. Robust vs Specific	★★★★★	Which HIs are universally vs conditionally useful?	Clear bimodal distribution in HI ranking	All HIs roughly equal = also publishable
3. Sup. vs Unsup.	★★★★☆	Can we skip SOH labels?	Gap < 1% in most scenarios	Large gaps guide calibration needs
4. Attention networks	★★★★☆	What multivariate interactions exist?	Physically interpretable patterns	Report raw heatmaps without over-interpretation
5. Size saturation	★★★★☆	How many HIs are enough?	Clear elbow per scenario	Move to supplementary
6. Cross-dataset	★★★★★	Do findings generalize?	RMSE < 3% on cross-dataset	Report gaps with analysis of why

7. Expected Contributions

Contribution 1: First systematic partial-discharge scenario classification framework for LFP batteries (dual manual + automatic classification with mutual validation).

Contribution 2: HI Efficacy Map — a scenario × HI performance matrix serving as a reusable community reference for BMS engineers and researchers.

Contribution 3: Robust HI vs. Scenario-Specific HI classification — practical guidelines for which features to monitor under different operating profiles.

Contribution 4: Cross-variate attention-based multivariate HI interaction analysis — first application of iTransformer attention as an interpretable analytical tool for battery feature analysis, surpassing univariate correlation methods.

Contribution 5: Supervised vs. unsupervised feature selection gap quantification per scenario — directly informing real-world deployability without SOH labels.

8. Risk Assessment and Mitigation

Risk	Severity	Mitigation
Optimal HI subset is the same across all scenarios	Medium	This is also a valid finding ('universal HI set exists'). Reframe contribution as confirming robustness.
iTransformer attention patterns are uninterpretable	Medium	RF feature importance serves as alternative. Report raw heatmaps; physical interpretation is bonus, not requirement.
iTransformer shows no advantage over RF/LSTM	Low	Model performance is secondary. Core contribution is the analysis framework and HI Efficacy Map, not model superiority.
Clustering produces unstable scenario boundaries	Medium	Manual scenarios (H/L/W/R) serve as stable fallback. Report both and compare.
Cross-dataset performance is poor	Medium	Analyze WHY (charging vs. discharge diversity). The gap analysis itself is a contribution.
LFP flat plateau limits HI discriminability in mid-SOC	High	This is expected for Scenario H. Document as a known limitation; it strengthens the case for scenario-specific analysis.

9. Development Roadmap

Week	Task	Deliverable	Dependencies
1–2	Data download, exploration, preprocessing pipeline	Clean data loaders for MIT and HUST	None
2–3	Scenario classification (manual + automatic)	Scenario labels + clustering visualization	Data loaders
3–4	SOC overlap algorithm + feature extraction pipeline	Feature pool (30 HIs) per scenario per cell	Scenario labels
4–5	Feature subset construction (6 methods × 3 axes)	All HI subsets defined and stored	Feature pool
5–6	Train/val/test split + data pipeline validation	PyTorch DataLoaders for all experiment configs	HI subsets
6–8	iTransformer implementation + training across all configs	Trained models + attention weights extracted	DataLoaders
7–9	RF + LSTM baseline training	Baseline results + RF feature importances	DataLoaders
9–11	Result matrix construction + all 6 analyses	HI Efficacy Map + all figures and tables	All model results
11–12	Cross-dataset validation experiments	Generalizability results	Trained models
12–14	Manuscript writing	Draft manuscript	All analyses
14–16	Internal review, revision, submission	Final manuscript	Draft

10. Technical Stack

Component	Specification
Language	Python 3.10+
Deep Learning	PyTorch 2.x, iTransformer (Liu et al. ICLR 2024 open-source code base)
ML Baselines	scikit-learn (RF, PCA, Laplacian Score), XGBoost (optional)
Feature Selection	scikit-learn (Pearson, Spearman), pymrmr or custom MI-based mRMR
Clustering	scikit-learn (K-means, GMM, silhouette score, BIC)
Visualization	matplotlib, seaborn, networkx (for attention graphs), UMAP/t-SNE
Experiment Tracking	Weights & Biases (wandb) or MLflow
Data Format	HDF5 or Parquet for efficient I/O of large cycle datasets
Hardware	Single GPU (RTX 3090 / A100) — no multi-GPU needed given model size ~30K params
Version Control	Git + GitHub with experiment config versioning

11. Evaluation Metrics

Metric	Definition	Role
RMSE (%)	$\sqrt{\text{mean}((\hat{y} - y)^2)} \times 100$	Primary performance metric
MAE (%)	$\text{mean}(\hat{y} - y) \times 100$	Secondary performance metric
MAPE (%)	$\text{mean}(\hat{y} - y / y) \times 100$	Relative error metric
R ²	$1 - \text{SS}_{\text{res}} / \text{SS}_{\text{tot}}$	Goodness of fit
Silhouette Score	Cluster separation quality (−1 to 1)	Scenario classification quality
BIC	Bayesian Information Criterion	GMM cluster number selection
Wasserstein Distance	Distribution distance between scenario feature sets	Scenario separation quantification